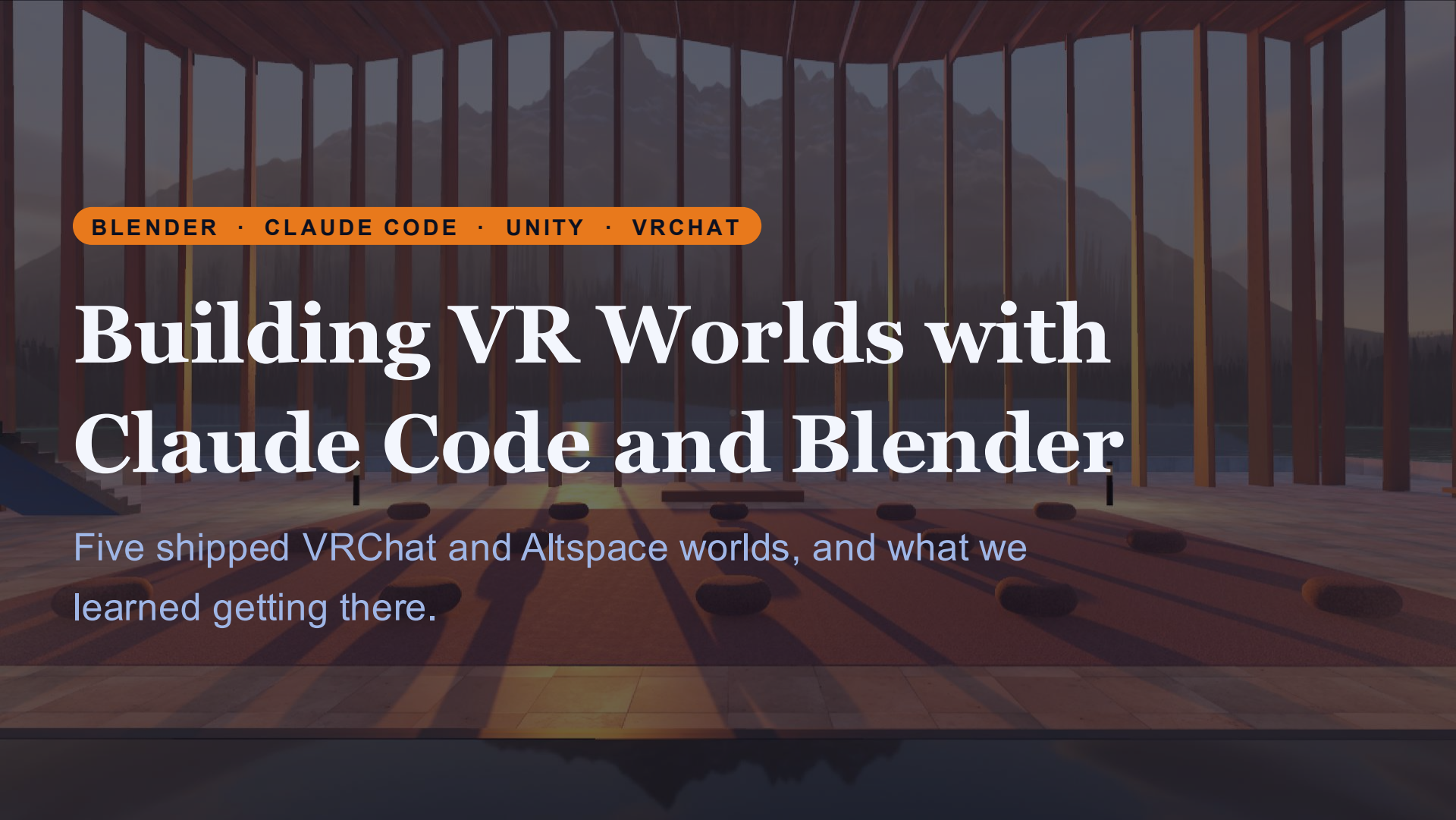


BLENDER · CLAUDE CODE · UNITY · VRCHAT

Building VR Worlds with Claude Code and Blender

Five shipped VRChat and Altspace worlds, and what we learned getting there.

THE LOOP

What is actually running

1. **Claude Code**
   drives the session and writes scripts
2. **Blender MCP**
   inspects and edits the live scene
3. **Python builders**
   prefix-delete, re-runnable, idempotent
4. **Headless verifiers**
   normals · walk · FBX · saved scene
5. **Unity → VRChat**
   atlas, colliders, lightmaps, upload

EVERY BUILD, HEADLESS

```
$ blender -b world.blend \
      -P build_garden.py

$ blender -b world.blend -P \
      normals_audit.py -- \
      --interior TUNNEL_ \
      --closed TOWER_shell

$ python verify_fbx.py \
      export/*.fbx
  UnitScaleFactor 100  OK
```

A world is a machine for generating confident wrong answers.

Almost every expensive bug passed every check that was running at the time. The checks were not lax — they were asking the wrong question.

And a wrong question returns a number, not an error.

SKILL.md, first paragraph

WORLD 01 · VRCHAT HOME WORLD

Space Station

A home world with one rule: load fast enough to feel instant, and still be worth staying in.

SCALE

24 m across

GEOMETRY

~16,000 triangles

BUILD SIZE

under 5 MB



Under 5 MB. Worlds of this kind are rarely close to that.

WORLD 01 · BUILD

Low-poly geometry, third-party interactivity



Lounge, with glass onto the galaxy skybox



Bar alcove — colour-block wall, no textures



Stepped seating under the star windows

Octagonal shell. Four alcoves — reading, bar, tech, tea — and four glass walls onto the galaxy skybox.

Built through the Blender MCP. Claude drives the live scene; the low-poly geometric style suits that loop.

Assembled in Unity. Video player, pens, beer pong, seating — sourced and configured, not modelled.

WORLD 02 · PROCEDURAL MATERIALS

Zen Garden

A walled Japanese garden — temple, dry raked sand, koi pond, arched bridge, stone lanterns.

SCALE

40 × 40 m

GEOMETRY

~115,000 triangles

BUILD SIZE

under 10 MB



Every material is a generator. Re-run the scripts and the whole garden retextures.

WORLD 02 · BUILD

Materials written as code, not painted



Dry garden — raked sand and stone lanterns



Koi pond and the arched wooden bridge



Temple at dusk, shoji lit by baked emission

No photographic textures. Wood, stone, sand, moss, water and foliage are all Python-written node graphs.

Retexturing touches no UVs. Tiling, scale, colour and wear are controlled in the script itself.

Shoji glow is baked emission, not translucency — so it renders identically on PC and on Quest.

WORLD 03 · NCC-1701 · TOS 60th

USS Enterprise

The sets that started it all: a 20 m bridge, a transporter room, and the corridor between them.

SCALE

20 m bridge

GEOMETRY

~225,000 triangles

MATERIALS

30 — only 2 textured



You arrive on the transporter pads, not in an empty lobby.

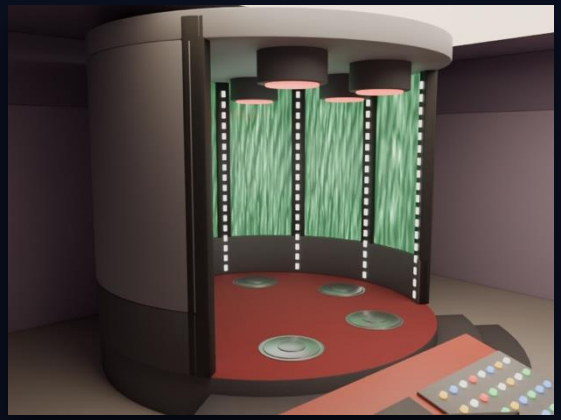
WORLD 03 · BUILD

400 parametric objects, almost no bitmaps



Bridge — viewscreen, helm and navigation

Blockout, then parametric fill. Roughly 400 objects on one shared palette of lavender, red and gold.



Transporter alcove; the moiré curtain is geometry

28 of 30 materials carry no texture. Button banks and the moiré curtain are geometry, not painted maps.



The corridor linking the two sets

Ten screens cost two textures. Public-domain NASA plates in two 1024² atlases with a JSON region map.

WORLD 04 · LARGEST WORLD · MYTH AS LAYOUT

Les Jardins de Perséphone

A French formal garden retelling the Persephone myth — spring west, underworld east, six sections.

SCALE

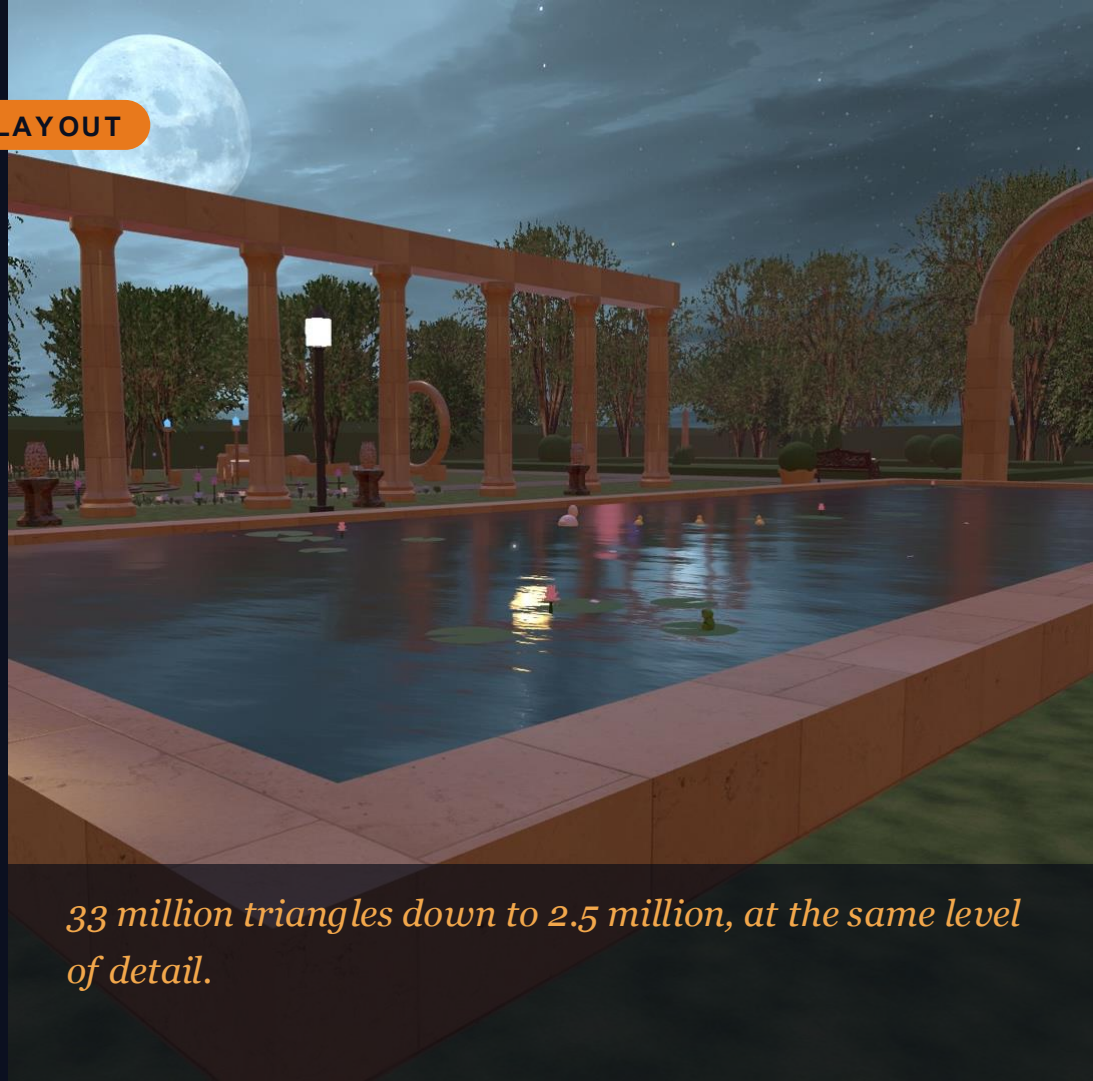
100 × 250 m

GEOMETRY

~2.5M triangles

SHIPS AS

2 FBX files



33 million triangles down to 2.5 million, at the same level of detail.

WORLD 04 · BUILD

A shared library, and its own exporter



Spring beds under the west arch



The central parterre on the axis



Pergola and geometry-node flower scatter

Every area is an idempotent function, rebuilt from constants — so the scene stays regenerable.

Trees became impostors. Sapling-grown, ortho-baked to crossed quads: 33M triangles down to 2.5M.

The exporter is part of the library: palette atlas, world-metre UVs, chunks, colliders, station markers.

WORLD 04 · HUNYUAN3D

Scripted from primitives, then generated



SCRIPTED 600 tris · flat



HUNYUAN3D 4,830 tris · PBR



SCRIPTED 5,786 tris · flat



HUNYUAN3D 10,726 tris · PBR

Nine frogs hide in the garden. The scripted frog was a handful of spheres; the generated one is a character worth finding.

Generated meshes still go through the pipeline. Both arrived with packed textures FBX can’t carry; one fixer now unpacks them to 1024² for Unity.

WORLD 05 · ONE CONSTANTS FILE

Lakeside Lounge

A meditation studio facing a glacial lake at sunset. Neuroaesthetics and Biophilic design.

SCALE

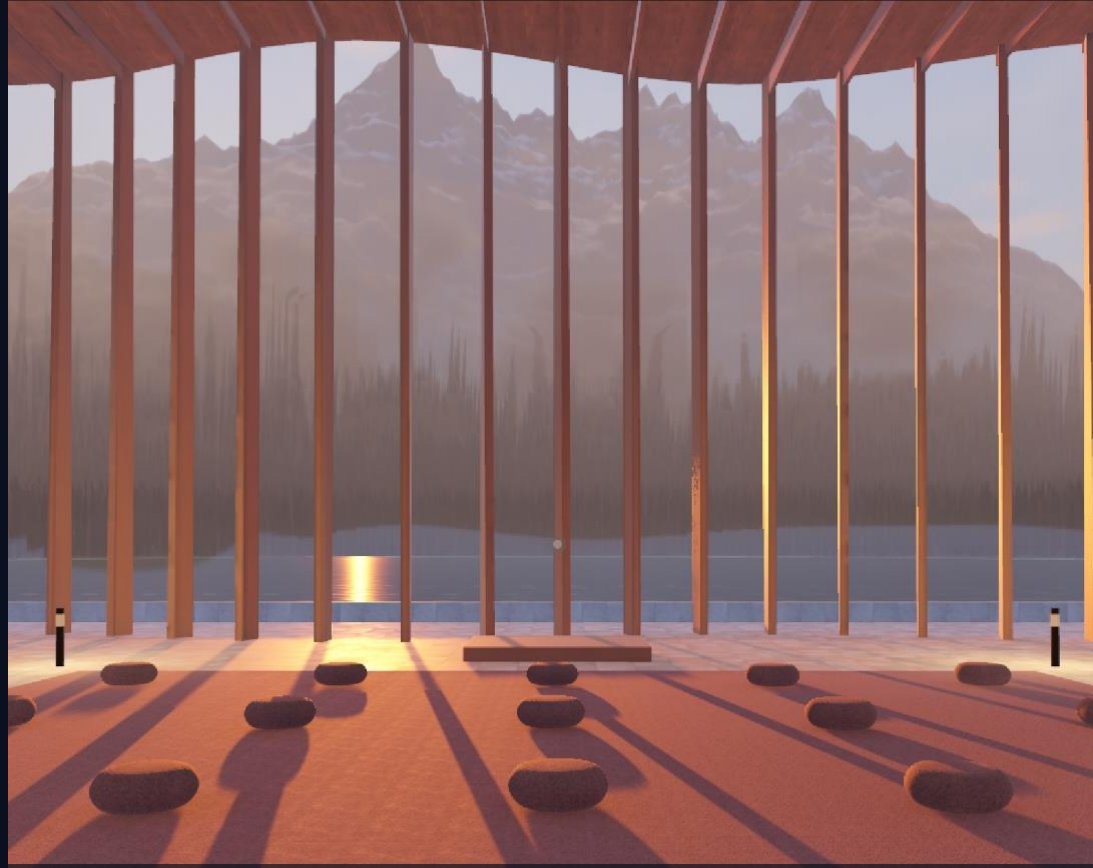
20 × 32 m, 12 m tall

GEOMETRY

~187,000 triangles

BACKDROP

one 8K panorama



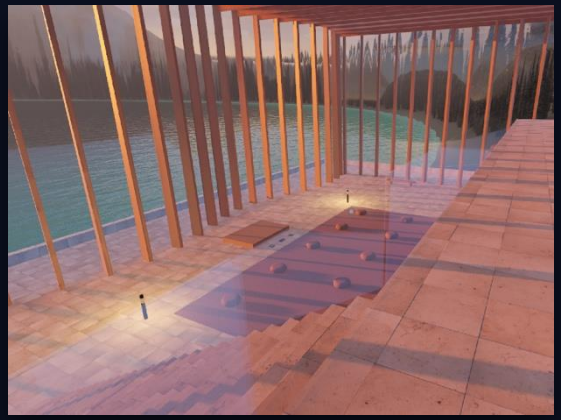
The mountains are a projected panorama. As geometry they ran past a billion triangles.

WORLD 05 · BUILD

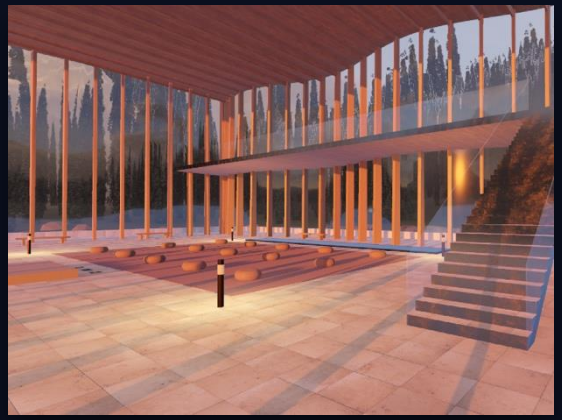
One sine curve drives the whole building



The curtain wall — 68 white-oak fins



Sunset raking through floor-to-ceiling glass

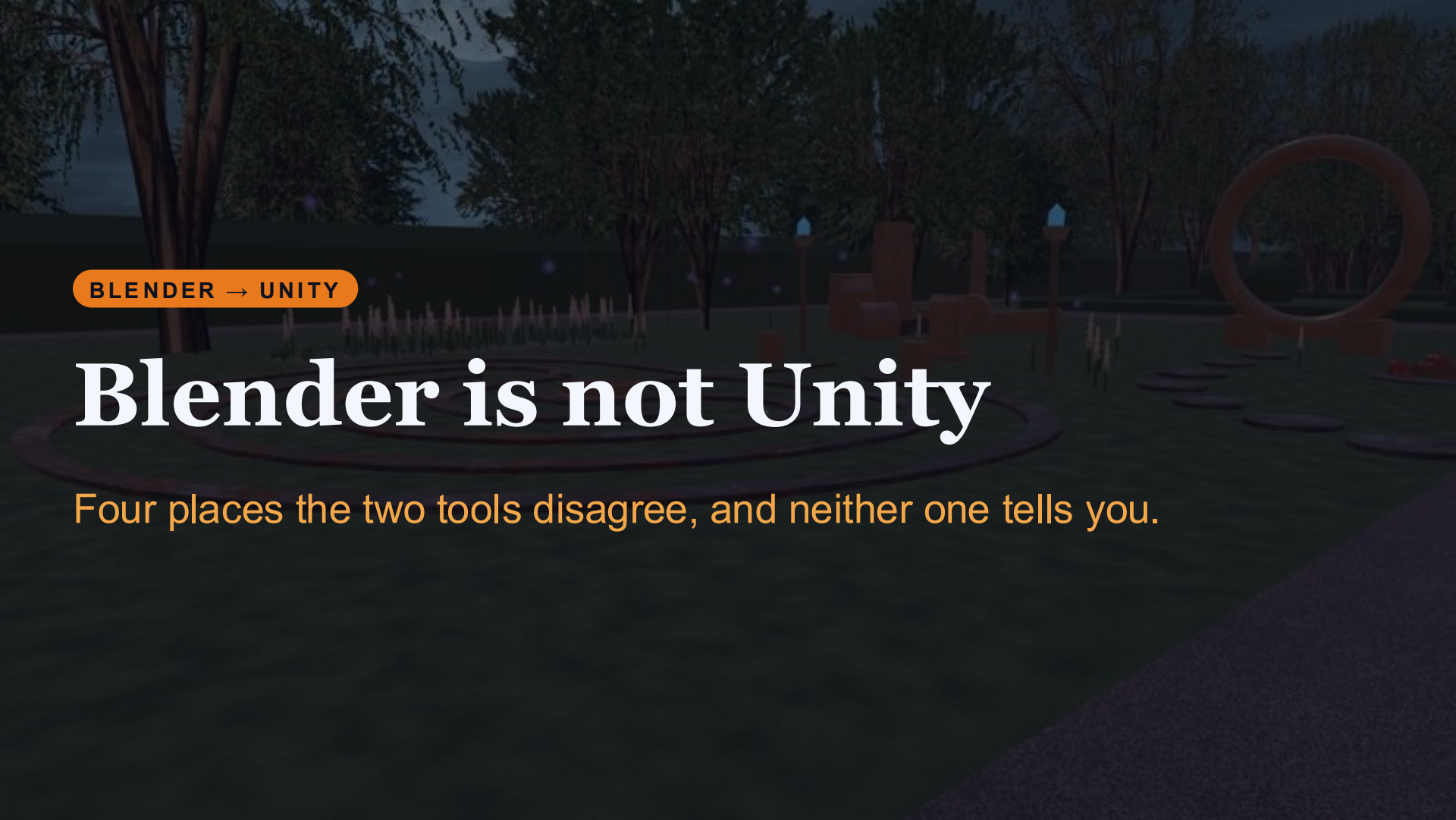


Ribbed oak ceiling over the practice floor

One constants file drives it. Wall bow, fin taper and ceiling ribs follow a single sine curve.

The valley is a skydome — an 8K Cycles panorama, because the real thing is a billion triangles.

Limestone, white oak and blackened steel against mauve wool felt, lit by a low sun through the glass.



BLENDER → UNITY

Blender is not Unity

Four places the two tools disagree, and neither one tells you.

DIFFERENCE 01 · MATERIALS

Procedural materials stay in Blender

IN BLENDER

Every surface is a node graph.

Cycles evaluates it in world space.

So the scene has zero UVs.

WHAT REACHES UNITY

one colour

per material. Neither URP nor Built-in can evaluate the graph.

UV unwrap and bake are an export stage, not polish.

Three ways out. Bake lighting only and keep albedo procedural in a custom shader. Bake to tileables plus a small palette. Or bake unique albedo per surface, which large terrain cannot afford.

DIFFERENCE 02 · CULLING

A face has one side in Unity

both

sides of a face are drawn in Blender's viewport, by default

back

faces are culled by every stock Unity shader

opt-in

two-sided rendering is set per material, in both tools

Make Blender show what Unity will. Turn on Backface Culling in viewport shading, or the Face Orientation overlay: blue is a front face, red is a back.

Two-sided is for cards, not for fixes. Impostors and foliage quads get a two-sided cutout material. Everything else gets its winding corrected at the source.

Nothing errors. The face is simply not drawn — hence Bug 01.

DIFFERENCE 03 · LIGHTS

A watt is not an intensity

watts are what Blender lights are set in: radiant power, with the sun in W/m². Unity's Built-in intensity is a unitless multiplier, with a Range cutoff Blender does not have.

P / 4π turns a point light's watts into radiant intensity in W/sr. It is a starting value for the few lights that stay realtime; the rest bake.

AgX grades every Blender render. Unity applies no view transform until you add a tonemapper, so AgX-tuned colours read darker and more saturated there.

Emission is a light too. In Blender it peaks near 1–2 before the view transform bleaches it white; in Unity, lamp glass and embers were pushed to 2–4 to read as lit.

DIFFERENCE 04 · TRANSFORMS

One model, three coordinate systems

	BLENDER	FBX FILE	UNITY
Up axis	Z	Y	Y
Handedness	right-handed	right-handed	left-handed
Unit	metre	centimetre	metre
Forward	−Y	−Z	+Z

Two conversions, and Blender’s exporter defaults put both on the object, not in the vertices. The file still renders at the right size and the right way up. The two flags that fix it are Bug 04; the handedness flip is Bug 05.



PART TWO

What went wrong

Eight bugs. Every one of them was green when it shipped.

BUG 01 · NORMALS

Blender is double-sided. Unity culls backfaces.

BLENDER VIEWPORT



The inverted face renders perfectly. It looks finished.

UNITY



The same face is invisible. Nothing about the mesh changed.

Your eyes in the Blender viewport cannot see this bug.

Standing instruction, not a task. Audit normals on everything generated or edited, every session, unasked.

Decide inversion or hole before flipping. They look identical in Unity and need opposite fixes.

BUG 01 · CONTINUED

Pick the test by topology — the wrong one lies

TOPOLOGY	TEST	INVERTED READS
Closed solid	per-island signed volume > 0, about its own centroid	negative
Open shell	area-weighted fraction of faces with world normal.z > 0	≈ 0.000
Interior void	dot(normal, axis − face_centre) > 0 — normals face the player	fraction < 0.5
Horizontal tube	ray test inward from outside the bounding sphere	frac_up is useless: 0.48–0.52 either way

Signed volume on an open mesh is meaningless — it produced 51 false positives in a single sweep.

scripts/normals_audit.py implements all four

BUG 02 · INSTRUMENTS

The verifier that lied

0.00 %
reported for the good sphere

100.00 %
reported for the flipped one

PASS
on a luminance of −431602100

Sign-inverted. The ray normals test had the comparison backwards, so it had already cleared an inverted mesh as clean.

Identical garbage compares equal. A headless lighting-leak test returned PASS because both frames rendered the same nonsense.

A verifier that has never flagged a known-bad input is not evidence of anything.

BUG 03 · THE HAND-OFF

Verify the destination, not the export

ALL GREEN

Lightmaps baked.

Atlases correct.

FBX passing a control-validated verifier.

IN UNITY

median 1.5

The world rendered essentially black.

`LightmapSettings.lightmaps does not survive a scene save.`

Probe the saved scene, not the build log. Ask the engine's own physics whether a route is walkable, and read every setting back off the asset after writing it.

BUG 04 · EXPORT

Blender's FBX defaults are wrong — twice

BOTH ARE REQUIRED, AND THEY ARE INDEPENDENT

```
apply_scale_options = 'FBX_SCALE_ALL'    # else UnitScaleFactor 1.0
                                         #   + Lcl Scaling [100,100,100]
bake_space_transform = True              # else Lcl Rotation [-90,0,0]
                                         #   over Z-up verts, Y-up header
```

Both still render at the right size. They surface weeks later as 100× child offsets, or a face-down model.

Verify the written file, never the exporter settings. UnitScaleFactor == 100 · UpAxis == 1 · neither fault signature · the bounding box carrying the height on Y.

Test the signatures, not any non-identity transform. Flagging every Lcl Scaling reported 564 legitimate tree transforms as faults.

scripts/verify_fbx.py parses the binary FBX

BUG 05 · COORDINATES

It is not a Z-up / Y-up swap. It flips handedness.

$$(x,\ y,\ z) \rightarrow (-x,\ z,\ -y) \qquad \det = -1$$

Writing it as (x, z, y) mirrors the world. Every hand-converted position lands on the far side of the map while the FBX geometry stays perfectly correct — so nothing in the file is wrong and nothing looks wrong. A camera placed that way rendered the world from behind.

Check any hand-converted coordinate against an object the FBX actually placed. That is the only comparison that can catch it.

BUG 06 · WALKABILITY

A width you fit through is not a width you walk along

PASSED FOR NINE SESSIONS

0.70 m a width a person fits through

THE REAL NUMBER

2.5–3 m a width they walk along while looking at the scenery

≤ 35°
slope, anything walked

≤ 0.30 m
riser — the CharacterController step

≥ 1.80 m
headroom

0.5 m
kerb on any exposed edge

Use a colliding walk, not a downward raycast. A downward sampler treats any hit as floor, so it walks ceilings — and returns a confident coverage number while it does.

BUG 07 · BAKING

The light budget bites at export

249 of 250 realtime lights were broad fill that bakes into the atlas and never ships as a light at all.

> 12× bake speedup from dropping volume links. 139 volumes per ray meant a 0.01 M-texel pilot did not finish in 25 minutes.

0.39181 what a known linear 0.5000 writes as to an AgX PNG (0.48310 to EXR). Re-baking to EXR is not the same image with more bits — it is an ungraded one.

Lightmap density follows lit need, not area: 8 texels/m within 20 m of a light, 0.5 elsewhere — what fits a 5.5 million m² world into two 4096² maps.

BUG 08 · REGRESSION

Fix at the source, or you will meet it again

world.blend — CONTENT MASTER

Live modifiers, procedural materials, the authored world. All content edits happen here.

export/world_export.blend

Disposable. Rebuilt from the master every run. Never hand-edited — a fix landed here is gone next regeneration.

One garden lantern was documented as fixed in the master. It was still broken there; the fix had only ever landed in a throwaway export. Never trust a "fixed" note without re-measuring the geometry.

Assert idempotency by building twice and diffing the object-name lists. Assert that no name contains a dot — a missing prefix in the delete list is invisible until .001 appears.

THE REAL OUTPUT

The losses became a skill

LOADED AUTOMATICALLY, BY DESCRIPTION

```
~/.claude/skills/
  vr-world-building/
    SKILL.md
    references/
      normals.md
      blender-authoring.md
      unity-vrchat.md
      unity-altspace.md
    scripts/
      normals_audit.py
      verify_fbx.py
```

Written from losses, not from docs. Nearly every line exists because a session was spent on it.

The description is the trigger. It lists the symptoms — "the model is invisible in Unity" — so it loads before the mistake repeats, not after.

The scripts gate the build. normals_audit.py validates each test against a control of its own topology, then exits non-zero.

WHAT IT DISTILS TO

The five habits

1. **Validate the instrument against a known-bad control.**
   If it has never screamed at something broken, it is not evidence.

2. **Survey before you design; check the premise arithmetic first.**
   One stated bridge span was geometrically impossible — a 60° wall, not a gap.

3. **Verify the destination, not the export.**
   Every Blender-side check can pass while nothing reaches the engine.

4. **Fix at the source — the content master and the builder code.**
   Anything patched in a disposable file regresses on the next run.

5. **Measure renders; do not eyeball them.**
   If an input change moves no pixels, that is evidence, not a null result.